25901397 MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER- Discipline Specific Core Course (DSC) - MG4DSCMBG200 -

ENVIRONMENTAL MICROBIOLOGY - (2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Part A - Short Notes Questions - Answer any 10 questions - Each question carries 2 marks. 2X10=20

1. What is liquid waste? [K] / [CO3]

- Waste in liquid form from domestic/industrial activities (1) Examples: sewage, wastewater (1)

2. What is an Estuary? [K] / [CO2]

- Semi-enclosed coastal water body (1) Where freshwater mixes with seawater (1)

3. List any two types of biogas plants.[K] / [CO3]

- Floating drum type (1) Fixed dome type (1)

4. Define gravity-based air sampling and name one device. [K] / [CO1]

- Collection of microorganisms by settling under gravity (1) Example: settle plate (Petri dish) (1)

5. What is an Oxidation pond? [K] / [CO2]

- Shallow pond used for wastewater treatment (1). Uses microorganisms and sunlight to degrade organic matter (1)

6. Disadvantages of ozone in water purification [K] / [CO2]

- Expensive process (1) No residual disinfectant effect (1)

7. One environmental factor affecting hydrocarbon degradation [K] / [CO3]

- Oxygen availability (1) Affects microbial activity (1)

8. What is the Pelagic zone? [K] / [CO2]

- Open water region of oceans/lakes (1) Away from bottom and shore (1)

9. What is vertical migration? [K] / [CO2]

- Movement of aquatic organisms up and down in water (1). Diurnal movement/example (1)

10. Phases of composting [U] / [CO3]

- Mesophilic phase (1), Thermophilic and maturation phase (1)

11. Function of biosurfactants in oil degradation [U] / [CO3]

- Reduce surface tension/emulsify oil (1). Increase availability of hydrocarbons to microbes (1)

12. What is Biofouling? [K] / [CO2]

- Accumulation of microorganisms on surfaces (1) Forms biofilms on pipes/ships (1)

Part B - Short Essay Questions - Answer any 4 questions - Each question carries 5 marks – 5X4=20

13. Mechanisms of microbial dispersion into air (from humans)

- Sneezing and coughing release droplets. Talking and breathing produce aerosols. Shedding of skin cells carrying microbes. Movement resuspends settled dust microbes. Poor ventilation increases airborne spread. [Any 5 points – 5 marks]

14. Zonation of mangroves based on salinity

- Euhaline, Polyhaline, Mesohaline, Oligohaline, Limnetic Zone [5]

15. Phytovolatilization for mercury remediation

- Plants absorb mercury from soil - Convert to a less toxic volatile form - Release it into the atmosphere through leaves - Reduce soil contamination - Eco-friendly and cost-effective method [5]

16. Penicillium and Aspergillus as indoor air indicators

- Common fungi in indoor environments - Grow well in damp and humid conditions -Presence indicates moisture problems - produce spores affecting air quality- Associated with allergies and health issues [Any 5 points -5 marks]

17. **Eutrophication**

- Excess nutrients (nitrogen, phosphorus) in water - Leads to algal bloom formation-Reduces oxygen levels in water - Causes death of aquatic organisms - due to sewage and fertiliser runoff [5]

18. Bioremediation of oil-contaminated coastal soil

- Use biostimulation (adding nutrients) - Enhances growth of native oil-degrading microbes - Microbes break down hydrocarbons – or any other- Cost-effective and environmentally safe. [5 Relevant points – 5 marks]

Part C - Essay Question - Answer any 1 question - Each question carries 10 marks. 10X1=10

19. Cometabolism in pesticide degradation (10 marks)

- Definition: degradation of pesticide without being used as an energy source (2). Requires the presence of another substrate (2) - Microbes produce enzymes for the primary substrate (2) - These enzymes act on the pesticide (2) - Leads to partial/complete breakdown of toxic compounds (1) - Important in the removal of persistent pesticides from the environment (1)

20. Water pollution and its sources (10 marks)

- Water pollution definition (2): Domestic sewage adds organic matter and microbes (2), Industrial effluents release toxic chemicals/heavy metals (2), Agricultural runoff adds fertilisers and pesticides (2). Other sources: oil spills, plastics, pathogens (1). Causes harm to aquatic life and human health (1)